RESC vs VESC — FOC core and observer

RESC v0.1 · STM32F405 · reference compared: VESC firmware `release_7_00` · October 2026

> **In short.** RESC and VESC share the same textbook building blocks: amplitude-invariant Clarke/Park, d/q PI current loop, SVPWM, nonlinear flux observer with PLL. The **implementation and the logic around them are RESC's own** (an independent audit found no copied FOC code). RESC keeps VESC **interfaces** (protocol, configuration format, meaning of gains) so VESC Tool, displays and controllers keep working. VESC currently has many more features.

1. Observer

Topic	VESC 7.00	RESC
Base method	Nonlinear flux observer (Lee / Ortega et al., 2010), several selectable variants	Same family, written as an **active-flux** observer (Boldea): magnitude constraint `λ + (Ld − Lq)·id`; one equation for non-salient and salient (IPM) motors
Gain meaning	`foc_observer_gain` ≈ γ, $\gamma\lambda^2$ ≈ 1000 rad/s after detection	**Same meaning, kept on purpose** so VESC Tool values stay valid
Discrete stability	Gain used as configured	Correction **clamped** ($\gamma\lambda^2 \cdot dt \le 0.2$, step ≤ 0.5): an excessive gain cannot make the update diverge
Voltage input	Applied voltage from the modulation	**Average of the two duty sets actually applied** during the sampled interval, minus a current-sign dead-time term; bridge off: measured phase voltages (flying start)
Speed output	PLL speed	**Derivative of the PLL angle** (`ω_int + kp·e`). With default PLL gains the integrator alone lagged ~30 % under acceleration in simulation
Health check	—	Estimated / expected flux ratio monitored: outside 0.5–2 for 20 ms → sync-loss fault
Variants, extras	Several observer types, low-speed gain, offset, R/λ temperature compensation, HFI	One observer, no temperature compensation, no HFI yet. Parallel observer comparison is planned but not implemented

2. Start-up and handover

Topic	VESC 7.00	RESC
Sensorless start	Open loop at `foc_openloop_rpm`, lock / ramp times, start current	**I/f**: current on the q axis of the open-loop frame, alignment, then speed ramp
Switch to observer	Speed thresholds	**Only when the observer is locked**: flux ratio 0.75–1.3 and observer speed within 25 % of open-loop speed, for 20 ms
Continuity	—	At handover, current references **and PI integrators are rotated** into the observer frame (back-EMF removed from integrators): no current or torque step
Failure	—	Timeout → start fault; below exit speed while running → back to I/f
Hall sensors	Table + interpolation	Edge angle = **circular midpoint** of sector centres; speed from same-direction edges; extrapolation **bounded to the current sector**
Hall ↔ observer	Transition around `foc_sl_erpm`	**Hysteresis switch** at 0.9 / 1.1 × `foc_sl_erpm` (no blending: small angle step possible if the Hall table is inaccurate)
Encoder	ABI, SPI, sin/cos	ABI only (timer in x4 mode)

3. Current loop

Topic	VESC 7.00	RESC
Structure	d/q PI, decoupling optional	d/q PI with decoupling and back-EMF feedforward **always on**, computed from **reference** currents (less noise)
Gains	`kp = L·bw`, `ki = R·bw`	**Same meaning**; computed automatically after R/L detection (bw = 1000 rad/s)
Voltage limit	Voltage limiting	Circle `m_max·Vbus/√3`, **d axis first**, q in the remaining voltage
Anti-windup	Integrator limiting	**Tracking**: on saturation the integrator is recomputed to match the saturated output
Delay compensation	Yes, in its own form	Inverse Park at `θ + ω·1.0·Ts`: new duty starts at the valley after the sample and is held one period
Duty mode	Duty / voltage control	**Always through the current loop** (PI on duty → iq): current limits stay active

4. Current sensing (3 low-side shunts)

Topic	VESC 7.00	RESC
ADC trigger	TIM8 slaved to TIM1	TIM1 OC4REF → TRGO, **centred** on the zero-vector window at the counter peak
Conversions	DMA regular group + injected currents	**Injected only**: 3 ADCs simultaneous; slow channels rotate on ranks 3–4
Phase validity	Modes with a fixed duty threshold in timer ticks	Per phase from the **actual low-side window time** vs `T_win`; rebuilt from 2 valid phases; predicted if fewer
Modulation	Max duty setting	Also **bounded so two phases are always measurable** in linear SVPWM
Fast overcurrent	Software check in the ISR	Software check + **ADC analog watchdog**, armed each period only on valid phases: cut ~1.5 µs after the sample, before the FOC computation

5. Motor identification (manual)

Measure	VESC 7.00	RESC
R	Current stepped until the resistive drop is large enough	Current ×1.5 until **≥ 1 V drop**, then **R = ΔV/ΔI between ½·I and I** (constant dead-time error cancels); warning if < 1 V
Ld / Lq	Own inductance measurement	**± voltage pulses on d, then q**, at a DC bias; only constant-voltage intervals used; adaptive amplitude
λ (flux)	Open-loop spin, then measurement	Ramp to a target duty; back-EMF **bridge on and bridge off**; rejected if they differ by more than 35 % (rotor slipped)
Hall / encoder	Own procedures	Slow rotation at fixed d current, 3 turns forward / back / forward, circular averages; encoder: ratio and direction, then offset
Auto wizard	Full automatic detection and configuration	**Disabled on purpose**: manual detections only, each followed by automatic computation of dependent gains

6. Status

Not yet in RESC (present in VESC)

Validation

- Field weakening, MTPA, HFI, position control
- R/λ temperature compensation, observer ↔ sensor blending
- Parallel observer comparison
- SPI and sin/cos encoders, dual motor

- **Simulation:** 45/45 closed-loop checks (motor, inverter, Ubox ADC model)
- **Real board:** USB, VESC Tool, real-time data, smooth rotation after flux detection
- **To validate on hardware:** new R/L procedure, higher power
- **Open:** no hardware-independent overcurrent cut on the Ubox board

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RESC — independent FOC firmware for STM32F405 boards compatible with VESC hardware · GPL v3 · github.com/Foujiwara/RESC · VESC reference: github.com/vedderb/bldc (release_7_00, GPL v3). Experimental power-electronics firmware; bench use with current-limited supply.